

Lab7

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(1) ① P is the class of languages that are decidable in polynomial time on deterministic single-tape Turing machine.

$$P = \bigcup \text{TIME}(n^k)$$

② NP is the class of languages that have polynomial time verifiers.

A verifier for a language A is an algorithm V,

$$A = \{ \langle w \rangle \mid V \text{ accepts } \langle w, c \rangle \text{ for some string } c \}, c \text{ is certificate.}$$

From theorem 7.20, a language is in NP iff it is decided by some non-deterministic polynomial time Turing machine.

(2) ① PATH Problem

$$\text{PATH} = \{ \langle G, s, t \rangle \mid G \text{ is a directed graph that has a directed path from } s \text{ to } t \}$$

② RELPRIME Problem

$$\text{RELPRIME} = \{ \langle x, y \rangle \mid x \text{ and } y \text{ are relatively prime if } 1 \text{ is the greatest integer that can divide both numbers} \}$$

③ SAT Problem

$$\text{SAT} = \{ \langle \phi \rangle \mid \phi \text{ is a satisfiable Boolean formula if the assignment of } 1 \text{ and } 0 \text{ to literals makes the value of formula is } 1 \}$$

④ HAMPATH problem

$$\text{HAMPATH} = \{ \langle G, s, t \rangle \mid G \text{ is a directed graph that has a Hamiltonian path from } s \text{ to } t \text{ and Hamiltonian path is a directed path that goes through each node exactly once} \}$$

⑤ COMPOSITES Problem

$$\text{COMPOSITES} = \{ \langle x \rangle \mid x = p \cdot q \text{ for integers } p, q > 1 \}$$

⑥ CLIQUE

$$\text{CLIQUE} = \{ \langle G, k \rangle \mid G \text{ is an undirected graph with } k\text{-clique and } k\text{-clique is that each } 2 \text{ of } k \text{ nodes are connected by an edge} \}$$

(3)

① Big-O notation

Let f and g be functions $f, g: \mathbb{N} \rightarrow \mathbb{R}^+$. $f(n) = O(g(n))$ if positive integers c and n_0 exist such that for every $n > n_0$, $f(n) \leq c \cdot g(n)$.

② Small-O notation

Let f and g be functions $f, g: \mathbb{N} \rightarrow \mathbb{R}^+$. $f(n) = o(g(n))$ if $\lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = 0$.

③ Verifier

A verifier for a language A is an algorithm V ,

$A = \{ \langle w \rangle \mid V \text{ accepts } \langle w, c \rangle \text{ for some string } c \}$,

c is certificate for additional extra verification information.

(4)

$M =$ "On input $\langle G, s, t \rangle$, where G is directed graph with node s and t :

1. place a mark on s
2. Repeat following programs until no additional nodes are marked:
Scan all edges of G . If edge (a, b) is found going from node a to an unmarked node b , mark node b .
3. If t is marked, accept. Otherwise, reject.

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(5)

$N =$ "On input $\langle G, k \rangle$ where G is a graph:

1. Nondeterministic select a subset c of k nodes of G .
2. Test whether G includes all edges connecting all nodes in c .
3. If yes, accept. Otherwise, reject.